4.3 The Electric Displacement

4.3.1 Gauss's Law in the Presence of Dielectrics

In Section 4.2 we found that the effect of polarization is to produce accumulations of (bound) charge, $\rho_b = -\boldsymbol{\nabla} \cdot \mathbf{P}$ within the dielectric and $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ on the surface. The field due to polarization of the medium is just the field of this bound charge. We are now ready to put it all together: the field attributable to bound charge plus the field due to everything *else* (which, for want of a better term, we call **free charge**, ρ_f). The free charge might consist of electrons on a conductor or ions embedded in the dielectric material or whatever; any charge, in other words, that is *not* associated with **P**. Within the dielectric, the total charge density can be written

$$\rho = \rho_b + \rho_f, \tag{4.20}$$

and Gauss's law reads

$$\epsilon_0 \boldsymbol{\nabla} \cdot \mathbf{E} = \rho = \rho_b + \rho_f = -\boldsymbol{\nabla} \cdot \mathbf{P} + \rho_f,$$

where **E** is now the *total* field, not just the part generated by polarization.

It is convenient to combine the two divergence terms:

$$\boldsymbol{\nabla} \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho_f.$$

The expression in parentheses, designated by the letter **D**,

$$\boxed{\mathbf{D} \equiv \epsilon_0 \mathbf{E} + \mathbf{P},} \tag{4.21}$$

is known as the **electric displacement**. In terms of **D**, Gauss's law reads

$$\boxed{\boldsymbol{\nabla} \cdot \mathbf{D} = \rho_f,} \tag{4.22}$$

or, in integral form,

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f_{\text{enc}}}, \tag{4.23}$$

where $Q_{f_{\text{enc}}}$ denotes the total *free* charge enclosed in the volume. This is a particularly useful way to express Gauss's law, in the context of dielectrics, because *it makes reference only to free charges*, and free charge is the stuff we control. Bound charge comes along for the ride: when we put the free charge in place, a certain polarization automatically ensues, by the mechanisms of Section 4.1, and this polarization produces the bound charge. In a typical problem, therefore, we know ρ_f, but we do not (initially) know ρ_b; Eq. 4.23 lets us go right to work with the information at hand. In particular, whenever the requisite symmetry is present, we can immediately calculate **D** by the standard Gauss's law methods.

Example 4.4. A long straight wire, carrying uniform line charge λ, is surrounded by rubber insulation out to a radius a (Fig. 4.17). Find the electric displacement.

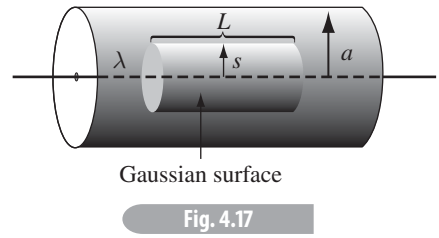


Fig. 4.17

Solution. Drawing a cylindrical Gaussian surface, of radius s and length L, and applying Eq. 4.23, we find

$$D(2\pi sL) = \lambda L.$$

Therefore,

$$\mathbf{D} = \frac{\lambda}{2\pi s}\hat{\mathbf{s}}. \tag{4.24}$$

Notice that this formula holds both within the insulation and outside it. In the latter region, $\mathbf{P} = \mathbf{0}$, so

$$\mathbf{E} = \frac{1}{\epsilon_0}\mathbf{D} = \frac{\lambda}{2\pi\epsilon_0 s}\hat{\mathbf{s}} \quad \text{for } s > a.$$

Inside the rubber, the electric field cannot be determined, since we do not know $\mathbf{P}$.

It may appear to you that I left out the surface bound charge σ_b in deriving Eq. 4.22, and in a sense that is true. We cannot apply Gauss's law precisely *at* the surface of a dielectric, for here ρ_b blows up,[7] taking the divergence of $\mathbf{E}$ with it. But everywhere *else* the logic is sound, and in fact if we picture the edge of the dielectric as having some finite thickness, within which the polarization tapers off to zero (probably a more realistic model than an abrupt cutoff anyway), then there *is* no surface bound charge; ρ_b varies rapidly but smoothly within this "skin," and Gauss's law can be safely applied *everywhere*. At any rate, the integral form (Eq. 4.23) is free from this "defect."

Problem 4.15. A thick spherical shell (inner radius a, outer radius b) is made of dielectric material with a "frozen-in" polarization

$$\mathbf{P}(\mathbf{r}) = \frac{k}{r}\hat{\mathbf{r}},$$

[7] The polarization drops abruptly to zero outside the material, so its *derivative* is a delta function (see Prob. 1.46). The surface bound charge *is* precisely this term – in this sense it is actually *included* in ρ_b, but we ordinarily prefer to handle it separately as σ_b.

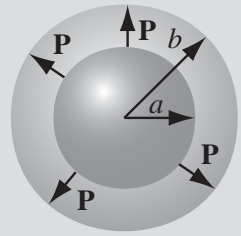


Fig. 4.18

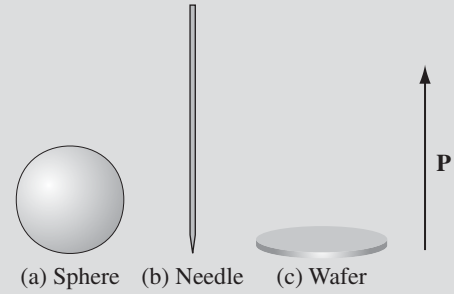


Fig. 4.19

where k is a constant and r is the distance from the center (Fig. 4.18). (There is no *free* charge in the problem.) Find the electric field in all three regions by two different methods:

(a) Locate all the bound charge, and use Gauss's law (Eq. 2.13) to calculate the field it produces.
(b) Use Eq. 4.23 to find $\mathbf{D}$, and then get $\mathbf{E}$ from Eq. 4.21. [Notice that the second method is much faster, and it avoids any explicit reference to the bound charges.]

Problem 4.16. Suppose the field inside a large piece of dielectric is $\mathbf{E}_0$, so that the electric displacement is $\mathbf{D}_0 = \epsilon_0 \mathbf{E}_0 + \mathbf{P}$.

(a) Now a small spherical cavity (Fig. 4.19a) is hollowed out of the material. Find the field at the center of the cavity in terms of $\mathbf{E}_0$ and $\mathbf{P}$. Also find the displacement at the center of the cavity in terms of $\mathbf{D}_0$ and $\mathbf{P}$. Assume the polarization is "frozen in," so it doesn't change when the cavity is excavated.
(b) Do the same for a long needle-shaped cavity running parallel to $\mathbf{P}$ (Fig. 4.19b).
(c) Do the same for a thin wafer-shaped cavity perpendicular to $\mathbf{P}$ (Fig. 4.19c).

Assume the cavities are small enough that $\mathbf{P}$, $\mathbf{E}_0$, and $\mathbf{D}_0$ are essentially uniform. [*Hint:* Carving out a cavity is the same as superimposing an object of the same shape but opposite polarization.]

4.3.2 A Deceptive Parallel

Equation 4.22 looks just like Gauss's law, only the *total* charge density ρ is replaced by the *free* charge density ρ_f, and $\mathbf{D}$ is substituted for $\epsilon_0 \mathbf{E}$. For this reason, you may be tempted to conclude that $\mathbf{D}$ is "just like" $\mathbf{E}$ (apart from the factor ϵ_0), except that its source is ρ_f instead of ρ: "To solve problems involving dielectrics, you just forget all about the bound charge – calculate the field as you ordinarily would, only call the answer $\mathbf{D}$ instead of $\mathbf{E}$." This reasoning is seductive, but the conclusion is false; in particular, there is no "Coulomb's law" for $\mathbf{D}$:

$$\mathbf{D}(\mathbf{r}) \neq \frac{1}{4\pi} \int \frac{\hat{\boldsymbol{\imath}}}{\imath^2} \rho_f(\mathbf{r}')\, d\tau'.$$

The parallel between $\mathbf{E}$ and $\mathbf{D}$ is more subtle than that.

For the divergence alone is insufficient to determine a vector field; you need to know the curl as well. One tends to forget this in the case of electrostatic fields because the curl of **E** is always zero. But the curl of **D** is *not* always zero:

$$\nabla \times \mathbf{D} = \epsilon_0(\nabla \times \mathbf{E}) + (\nabla \times \mathbf{P}) = \nabla \times \mathbf{P}, \tag{4.25}$$

and there is no reason, in general, to suppose that the curl of **P** vanishes. Sometimes it does, as in Ex. 4.4 and Prob. 4.15, but more often it does not. The bar electret of Prob. 4.11 is a case in point: here there is no free charge anywhere, so if you really believe that the only source of **D** is ρ_f, you will be forced to conclude that $\mathbf{D} = \mathbf{0}$ everywhere, and hence that $\mathbf{E} = (-1/\epsilon_0)\mathbf{P}$ inside and $\mathbf{E} = \mathbf{0}$ outside the electret, which is obviously wrong. (I leave it for you to find the place where $\nabla \times \mathbf{P} \neq \mathbf{0}$ in this problem.) Because $\nabla \times \mathbf{D} \neq \mathbf{0}$, moreover, **D** cannot be expressed as the gradient of a scalar – there is no "potential" for **D**.

Advice: When you are asked to compute the electric displacement, first look for symmetry. If the problem exhibits spherical, cylindrical, or plane symmetry, then you can get **D** directly from Eq. 4.23 by the usual Gauss's law methods. (Evidently, in such cases $\nabla \times \mathbf{P}$ is automatically zero, but since symmetry alone dictates the answer, you're not really obliged to worry about the curl.) If the requisite symmetry is absent, you'll have to think of another approach, and, in particular, you must *not* assume that **D** is determined exclusively by the free charge.

4.3.3 Boundary Conditions

The electrostatic boundary conditions of Section 2.3.5 can be recast in terms of **D**. Equation 4.23 tells us the discontinuity in the component perpendicular to an interface:

$$D^{\perp}_{\text{above}} - D^{\perp}_{\text{below}} = \sigma_f, \tag{4.26}$$

while Eq. 4.25 gives the discontinuity in parallel components:

$$\mathbf{D}^{\parallel}_{\text{above}} - \mathbf{D}^{\parallel}_{\text{below}} = \mathbf{P}^{\parallel}_{\text{above}} - \mathbf{P}^{\parallel}_{\text{below}}. \tag{4.27}$$

In the presence of dielectrics, these are sometimes more useful than the corresponding boundary conditions on **E** (Eqs. 2.31 and 2.32):

$$E^{\perp}_{\text{above}} - E^{\perp}_{\text{below}} = \frac{1}{\epsilon_0}\sigma, \tag{4.28}$$

and

$$\mathbf{E}^{\parallel}_{\text{above}} - \mathbf{E}^{\parallel}_{\text{below}} = \mathbf{0}. \tag{4.29}$$

You might try applying them, for example, to Probs. 4.16 and 4.17.

Problem 4.17. For the bar electret of Prob. 4.11, make three careful sketches: one of **P**, one of **E**, and one of **D**. Assume L is about $2a$. [*Hint:* **E** lines terminate on charges; **D** lines terminate on *free* charges.]

4.3.4 The Crystal Ambiguity

I began this chapter[8] with a description of dielectrics as neutral atoms or molecules that can be stretched or rotated by an electric field to form a collection of microscopic aligned dipoles. But what about an ionic crystal, such as ordinary table salt? It's a matrix of positive sodium ions (Na^+) and negative chlorine ions (Cl^-), as indicated schematically in Fig. 4.20. It's not a *conductor* (the electrons are attached to particular atoms); it behaves, in fact, as a kind of dielectric. But it does raise an intriguing puzzle: *What is its polarization,* **P***?*

You see, it's not clear *which* chlorine ion should be associated with a given sodium ion, to make a dipole **p** – the one above, or the one below, or the one to the left, or the one to the right? If you pick the ones below (Fig. 4.21a), then all the dipoles will be pointing upward, if you pick the ones above, they're all pointing downward (Fig. 4.21b); pick the ones to the left, and the dipoles all point to the right (Fig. 4.21c), pick the ones to the right, and the dipoles all point to the left (Fig. 4.21d). So **P** could point up, down, left, or right (or for that matter in, or out), depending on how you decide to form the dipoles! In each case there will be some leftover charges at the edges,

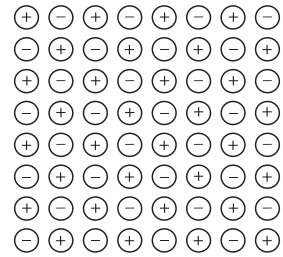


Fig. 4.20

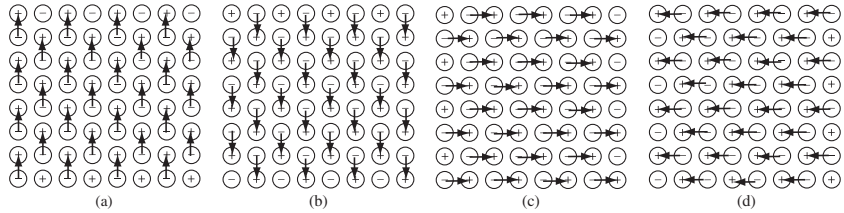


Fig. 4.21

[8] This section can be skipped without loss of continuity.

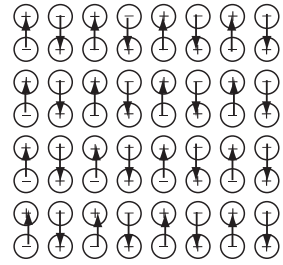


Fig. 4.22

which (since they are not associated with the basic dipoles) we identify as "free" charge, σ_f. It exactly cancels the "bound" surface charge, σ_b, in each case. But where it is located, and what sign it carries, depends (again) on our arbitrary choice. So not only is the polarization ambiguous, but so too is the distinction between free and bound charge.

We could even arrange for *zero* polarization (Fig. 4.22), with no free *or* bound charge. Indeed, in some sense this is the most "natural" choice. But in truth all of these different arrangements are equally "correct"; they all represent the same physical configuration. What we have discovered is that (at least in the case of ionic crystals), the notions of "dipole moment per unit volume," "free charge," and "bound charge" (and therefore also the electric displacement **D**) are intrinsically ill defined.

But it doesn't really matter. What matters is the *total* charge ρ. In the salt crystal (Fig. 4.20) everybody agrees that the total (macroscopic) charge density is zero, and the (macroscopic) electric field vanishes; they argue only about how to chop the charge into free and bound parts, and how to combine the ions to form dipoles. At the end of the day, it is *changes* in polarization (when, for example, you turn on a field and the ions adjust slightly in response) that carry genuine significance. This is where the physics resides, not in the story we tell ourselves to help us visualize what is happening.

4.4 Linear Dielectrics

4.4.1 Susceptibility, Permittivity, Dielectric Constant

In Sections 4.2 and 4.3 we did not commit ourselves as to the *cause* of **P**; we dealt only with the *effects* of polarization. From the qualitative discussion of Section 4.1, though, we know that the polarization of a dielectric ordinarily results from an applied electric field, which lines up the atomic or molecular dipoles. For many substances, in fact, the polarization is *proportional* to the field, provided **E** is not too strong:[9]

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}. \tag{4.30}$$

[9] Notice that Eq. 4.30 removes any ambiguity in the case of crystal dielectrics (Section 4.3.4).

The constant of proportionality, χ_e, is called the **electric susceptibility** of the medium (a factor of ϵ_0 has been extracted to make χ_e dimensionless). The value of χ_e depends on the microscopic structure of the substance in question (and also on external conditions such as temperature). I shall call materials that obey Eq. 4.30 **linear dielectrics**.[10]

Note that **E** in Eq. 4.30 is the *total* field; it may be due in part to free charges and in part to the polarization itself. If, for instance, we put a piece of dielectric into an external field $\mathbf{E}_0$, we cannot compute **P** directly from Eq. 4.30; the external field will polarize the material, and this polarization will produce its own field, which then contributes to the total field, and this in turn modifies the polarization, which .... Breaking out of this infinite regress is not always easy. You'll see some examples in a moment. The simplest approach is to begin with the *displacement*, at least in those cases where **D** can be deduced directly from the free charge distribution.

In linear media we have

$$\mathbf{D} = \epsilon_0\mathbf{E} + \mathbf{P} = \epsilon_0\mathbf{E} + \epsilon_0\chi_e\mathbf{E} = \epsilon_0(1 + \chi_e)\mathbf{E}, \tag{4.31}$$

so **D** is *also* proportional to **E**:

$$\mathbf{D} = \epsilon\mathbf{E}, \tag{4.32}$$

where

$$\epsilon \equiv \epsilon_0(1 + \chi_e). \tag{4.33}$$

This new constant ϵ is called the **permittivity** of the material. (In vacuum, where there is no matter to polarize, the susceptibility is zero, and the permittivity is ϵ_0. That's why ϵ_0 is called the **permittivity of free space**. I dislike the term, for it suggests that the vacuum is just a special kind of linear dielectric, in which the permittivity happens to have the value $8.85 \times 10^{-12}\,\mathrm{C^2/Nm^2}$.) If you remove a factor of ϵ_0, the remaining dimensionless quantity,

$$\epsilon_r \equiv 1 + \chi_e = \frac{\epsilon}{\epsilon_0}, \tag{4.34}$$

is called the **relative permittivity**, or **dielectric constant**, of the material. Dielectric constants for some common substances are listed in Table 4.2. (Notice that ϵ_r is greater than 1, for all ordinary materials; this just reflects the fact that polarization is typically in the same direction as the field.) Of course, the permittivity and the dielectric constant do not convey any information that was not already available in the susceptibility, nor is there anything essentially new in Eq. 4.32; the *physics* of linear dielectrics is all contained in Eq. 4.30.[11]

[10] In modern optical applications especially, *non*linear materials have become increasingly important. For these there is a second term in the formula for **P** as a function of **E** – typically a *cubic* term. In general, Eq. 4.30 can be regarded as the first (nonzero) term in the Taylor expansion of **P** in powers of **E**.

[11] As long as we are engaged in this excess of unnecessary terminology and notation, I might as well mention that formulas for **D** in terms of **E** (Eq. 4.32, in the case of linear dielectrics) are called **constitutive relations**.

Table 4.2. Dielectric constants (unless otherwise specified, values given are for 1 atm, 20°C). Data from *Handbook of Chemistry and Physics*, 91st ed., CRC Press, Boca Raton (2010).

Material	Dielectric constant	Material	Dielectric constant
Vacuum	1	Benzene	2.28
Helium	1.000065	Diamond	5.7–5.9
Neon	1.00013	Sodium chloride	5.9
Hydrogen (H_2)	1.000254	Silicon	11.7
Argon	1.000517	Methanol	33.0
Air (dry)	1.000536	Water	80.1
Nitrogen (N_2)	1.000548	Ice (−30°C)	104
Water vapor (100°C)	1.00589	$KTaNbO_3$ (0°C)	34,000

Example 4.5. A metal sphere of radius a carries a charge Q (Fig. 4.23). It is surrounded, out to radius b, by linear dielectric material of permittivity ϵ. Find the potential at the center (relative to infinity).

Solution. To compute V, we need to know $\mathbf{E}$; to find $\mathbf{E}$, we might first try to locate the bound charge; we could get the bound charge from $\mathbf{P}$, but we can't calculate $\mathbf{P}$ unless we already know $\mathbf{E}$ (Eq. 4.30). We seem to be in a bind. What we *do* know is the *free* charge Q, and fortunately the arrangement is spherically symmetric, so let's begin by calculating $\mathbf{D}$, using Eq. 4.23:

$$\mathbf{D} = \frac{Q}{4\pi r^2}\hat{\mathbf{r}} \quad \text{for all points } r > a.$$

(Inside the metal sphere, of course, $\mathbf{E} = \mathbf{P} = \mathbf{D} = \mathbf{0}$.) Once we know $\mathbf{D}$, it is a trivial matter to obtain $\mathbf{E}$, using Eq. 4.32:

$$\mathbf{E} = \begin{cases} \dfrac{Q}{4\pi\epsilon r^2}\hat{\mathbf{r}} & \text{for } a < r < b, \\ \dfrac{Q}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}} & \text{for } r > b. \end{cases}$$

The potential at the center is therefore

$$V = -\int_\infty^0 \mathbf{E}\cdot d\mathbf{l} = -\int_\infty^b \left(\frac{Q}{4\pi\epsilon_0 r^2}\right) dr - \int_b^a \left(\frac{Q}{4\pi\epsilon r^2}\right) dr - \int_a^0 (0)\, dr$$
$$= \frac{Q}{4\pi}\left(\frac{1}{\epsilon_0 b} + \frac{1}{\epsilon a} - \frac{1}{\epsilon b}\right).$$

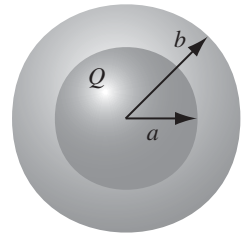


Fig. 4.23

As it turns out, it was not necessary for us to compute the polarization or the bound charge explicitly, though this can easily be done:

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E} = \frac{\epsilon_0 \chi_e Q}{4\pi\epsilon r^2}\hat{\mathbf{r}},$$

in the dielectric, and hence

$$\rho_b = -\boldsymbol{\nabla}\cdot\mathbf{P} = 0,$$

while

$$\sigma_b = \mathbf{P}\cdot\hat{\mathbf{n}} = \begin{cases} \dfrac{\epsilon_0\chi_e Q}{4\pi\epsilon b^2} & \text{at the outer surface,} \\ \dfrac{-\epsilon_0\chi_e Q}{4\pi\epsilon a^2} & \text{at the inner surface.} \end{cases}$$

Notice that the surface bound charge at a is *negative* ($\hat{\mathbf{n}}$ points outward *with respect to the dielectric*, which is $+\hat{\mathbf{r}}$ at b but $-\hat{\mathbf{r}}$ at a). This is natural, since the charge on the metal sphere attracts its opposite in all the dielectric molecules. It is this layer of negative charge that reduces the field, within the dielectric, from $1/4\pi\epsilon_0(Q/r^2)\hat{\mathbf{r}}$ to $1/4\pi\epsilon(Q/r^2)\hat{\mathbf{r}}$. In this respect, a dielectric is rather like an imperfect conductor: on a *conducting* shell the induced surface charge would be such as to cancel the field of Q *completely* in the region $a < r < b$; the dielectric does the best it can, but the cancelation is only partial.

You might suppose that linear dielectrics escape the defect in the parallel between **E** and **D**. Since **P** and **D** are now proportional to **E**, does it not follow that their curls, like **E**'s, must vanish? Unfortunately, it does *not*, for the line integral of **P** around a closed path that *straddles the boundary between one type of material and another* need not be zero, even though the integral of **E** around the same loop *must* be. The reason is that the proportionality factor $\epsilon_0\chi_e$ is different on the two sides. For instance, at the interface between a polarized dielectric and the vacuum (Fig. 4.24), **P** is zero on one side but not on the other. Around this loop $\oint \mathbf{P}\cdot d\mathbf{l} \neq 0$, and hence, by Stokes' theorem, the curl of **P** cannot vanish everywhere within the loop (in fact, it is *infinite* at the boundary).[12]

Of course, if the space is *entirely* filled with a homogeneous[13] linear dielectric, then this objection is void; in this rather special circumstance,

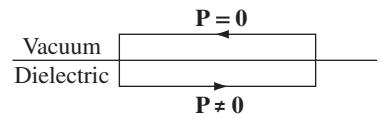


Fig. 4.24

[12] Putting that argument in differential form, Eq. 4.30 and product rule 7 yield $\boldsymbol{\nabla}\times\mathbf{P} = -\epsilon_0\mathbf{E}\times(\boldsymbol{\nabla}\chi_e)$, so the problem arises when $\boldsymbol{\nabla}\chi_e$ is not parallel to **E**.

[13] A **homogeneous** medium is one whose properties (in this case the susceptibility) do not vary with position.

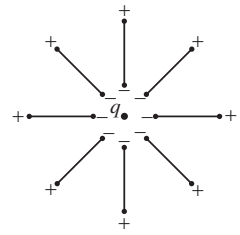


Fig. 4.25

$$\boldsymbol{\nabla} \cdot \mathbf{D} = \rho_f \quad \text{and} \quad \boldsymbol{\nabla} \times \mathbf{D} = \mathbf{0},$$

so $\mathbf{D}$ can be found from the free charge just as though the dielectric were not there:

$$\mathbf{D} = \epsilon_0 \mathbf{E}_{\text{vac}},$$

where $\mathbf{E}_{\text{vac}}$ is the field the same free charge distribution would produce in the absence of any dielectric. According to Eqs. 4.32 and 4.34, therefore,

$$\mathbf{E} = \frac{1}{\epsilon}\mathbf{D} = \frac{1}{\epsilon_r}\mathbf{E}_{\text{vac}}. \tag{4.35}$$

Conclusion: When all space is filled with a homogeneous linear dielectric, the field everywhere is simply reduced by a factor of one over the dielectric constant. (Actually, it is not necessary for the dielectric to fill *all* space: in regions where the field is zero anyway, it can hardly matter whether the dielectric is present or not, since there's no polarization in any event.)

For example, if a free charge q is embedded in a large dielectric, the field it produces is

$$\mathbf{E} = \frac{1}{4\pi\epsilon}\frac{q}{r^2}\hat{\mathbf{r}} \tag{4.36}$$

(that's ϵ, not ϵ_0), and the force it exerts on nearby charges is reduced accordingly. But it's not that there is anything wrong with Coulomb's law; rather, the polarization of the medium partially "shields" the charge, by surrounding it with bound charge of the opposite sign (Fig. 4.25).[14]

Example 4.6. A parallel-plate capacitor (Fig. 4.26) is filled with insulating material of dielectric constant ϵ_r. What effect does this have on its capacitance?

Solution. Since the field is confined to the space between the plates, the dielectric will reduce $\mathbf{E}$, and hence also the potential difference V,

[14] In *quantum* electrodynamics, the vacuum itself can be polarized, and this means that the effective (or "renormalized") charge of the electron, as you might measure it in the laboratory, is not its true ("bare") value, and in fact depends slightly on how far away you are!